

owner In file permissions, the userid that owns and controls access to a file.

owner rights In file permissions, the userid that owns and controls access to a file.

page address extension (PAE) Intel IA-32 CPU architecture hardware that allows 32-bit processors to access physical address space larger than 4GB.

page allocator The kernel routine responsible for allocating frames of physical memory.

page cache In file I/O, a cache that uses virtual memory techniques to cache file data as pages rather than file-system-oriented blocks for efficiency.

page directory In Intel IA-32 CPU architecture, the outermost page table.

page directory pointer table PAE pointer to page tables.

page fault A fault resulting from a reference to a non-memory-resident page of memory.

page frame A Windows virtual memory data structure.

page frame number (PFN) In Windows, the name of the indicator of the page frame address.

page number Part of a memory address generated by the CPU in a system using paged memory; an index into the page table.

page offset Part of a memory address generated by the CPU in a system using paged memory; the offset of the location within the page of the word being addressed.

page replacement In virtual memory, the selection of a frame of physical memory to be replaced when a new page is allocated.

page slot In Linux swap-space management, a part of the data structure tracking swap-space use.

page table In paged memory, a table containing the base address of each frame of physical memory, indexed by the logical page number.

page-directory entry (PDE) A Windows virtual-memory data structure.

page-fault frequency The frequency of page faults.

page-fault rate A measure of how often a page fault occurs per memory access attempt.

pageout policy Generally, an algorithm for deciding which memory pages to page out. In Linux, the virtual memory pageout policy uses a modified version of the second-chance algorithm.

pager The operating-system component that handles paging.

page-replacement algorithm In memory management, the algorithm that chooses which victim frame of physical memory will be replaced by a needed new frame of data.

page A fixed-sized block of logical memory.

page-table base register In paged memory, the CPU register pointing to the in-memory page table.

page-table entry (PTE) A Windows virtual memory data structure.

page-table length register A CPU register indicating the size of the page table.

paging A common memory management scheme that avoids external fragmentation by splitting physical memory into fixed-sized frames and logical memory into blocks of the same size called pages.

paging file The Windows term for backing store.

paging mechanism In Linux, the kernel component that carries out the transfer of pages back and forth to backing store.

paired password In authentication, a challenge-response set of secret keys, where only the correct response to the challenge provides authentication.

parallel file system (PFS) A file system that is LAN-based and treats N systems storing data and Y clients accessing the data as a single client-server instance; more complex than a client-server DFS but less complex than a cluster-based DFS. GPFS and Lustre are examples.

parallel regions Blocks of code that may run in parallel.

parallel systems Systems that have two or more hardware processors (CPU cores) in close communication, sharing the computer bus and sometimes the clock, memory, and peripheral devices.

parallelization The process of dividing a program into separate components that run in parallel on individual cores in a computer or computers in a cluster.

paravirtualization A technique in which a guest operating system is modified to work in cooperation with a virtual machine manager.

parent In a tree data structure, a node that has one or more nodes connected below it.

partition Logical segregation of storage space into multiple area; e.g., on HDDs, creating several groups of contiguous cylinders from the devices' full set of cylinders.

partition boot sector The NTFS boot control block.

passphrase A longer, generally more secure password composed of multiple words.

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password A secret key, usually used to authenticate a user to a computer.

path name The file-system name for a file, which contains all the mount-point and directory-entry information needed to locate the file (e.g., "C:/foo/bar.txt" and "/foo/bar.txt")

path-name translation The parsing of a file name into separate directory entries, or components.

PCIe bus A common computer I/O bus connecting the CPU to I/O devices.

peer-to-peer (p2p) A mode of distributed computing in which all nodes act as both clients of other nodes and servers to other nodes.

penetration test The scanning of a target entity to look for known vulnerabilities.

performance tuning The activity of improving performance by removing bottlenecks.

periodic A type of real-time process that repeatedly moves at fixed intervals between two modes: needing CPU time and not needing CPU time.

permissions An entity's access rights to an object (e.g., a user's access rights to a file).

per-process open-file table A kernel in-memory per-process data structure containing pointers to the system-wide open-file table, as well as other information, for all files the process has open.

personal firewall A software layer, either part of the operating system or added to a computer, limiting communication to and from a given host.

personal identification number A usually short and not very secure password composed of some combination of digits 0-9.

personal-area network (PAN) A network linking devices within several feet of each other (e.g., on a person).

petabyte (PB) $1,024^5$ bytes.

Peterson's solution A historically interesting algorithm for implementing critical sections.

phishing A class of social engineering attacks in which a legitimate-looking e-mail or website tricks a user into breaching confidentiality or enabling privilege escalation.

PHY The physical hardware component that connects to a network (implements layer 1 in the OSI model).

physical address Actual location in physical memory of code or data.

physical address space The set of all physical addresses generated by a program.

physical formatting The initialization of a storage medium in preparation for its use as a computer storage device.

physical-to-virtual (P-to-V) In virtualization, the conversion of a physical system's operating system and applications to a virtual machine.

Pico In WSL, a special Linux-enabling process that translates Linux system calls to the LXC and LXSS services.

pinning In memory management, locking pages into memory to prevent them from being paged out.

pipe A logical conduit allowing two processes to communicate.

platform as a service (PaaS) A software stack ready for application use via the Internet (e.g., a database server).

platter An HDD component that has a magnetic media layer for holding charges.

plug-and-play (PnP) manager In Windows, the manager responsible for detecting and enumerating devices when the system is booting and adding and removing devices when the system is running.

pluggable authentication module (PAM) A shared library that can be used by any system component to authenticate users.

plug-in An add-on functionality that expands the primary functionality of a process (e.g., a web browser plug-in that displays a type of content different from what the browser can natively handle).

point-to-point tunneling protocol (PPTP) A protocol in Windows and other systems allowing communication between remote-access server modules and client systems connected across a WAN.

policy A rule that defines what will be done.

policy algorithm In Linux, a part of the paging system that decides which pages to write out to backing store and when to write them.

polling An I/O loop in which an I/O thread continuously reads status information waiting for I/O to complete.

pool In virtual memory, a group of free pages kept available for rapid allocation (e.g., for copy-on-write). In ZFS, drives, partitions, or RAID sets that can contain one or more file systems.

pop The action of removing an item from a stack data structure.

port A communication address; a system may have one IP address for network connections but many ports, each for a separate communication. In computer I/O, a connection point for devices to attach to computers. In software development, to move code from its current platform to another platform (e.g., between operating systems or hardware systems). In the Mach OS, a mailbox for communication.

port driver In Windows I/O, the common driver for a class of devices.

port number In TCP/IP and UDP/IP networking, an address of a service on a system.

port set A collection of ports, as declared by a task, that can be grouped together and treated as one port for the purposes of the task.

portable An aspect of software that describes its ease of transfer between CPU architectures and computer systems.

portable executable (PE) The Windows format for executable files.

portals Gateways between requestors and services running on provider computers.

position-independent code (PIC) In Linux, binary code compiled from shared libraries that can be loaded anywhere in memory.

positioning time On an HDD, the time it takes the read-write head to position over the desired track.

power manager In Windows, the component that implements power management policies.

power users Users with unusually deep knowledge of a system.

power-of-2 allocator In the buddy system, an allocator that satisfies memory requests, in units sized as a power of 2, from a fixed-sized segment consisting of contiguous pages.

power-on self-test (POST) A firmware routine run at system power-on that tests the system for hardware issues, identifies and initializes many of the attached devices, and builds the description of the devices used by the advanced configuration and power interface (ACPI).

preemptive A form of scheduling in which processes or threads are involuntarily moved from the running state (e.g., by a timer signaling the kernel to allow the next thread to run).

preemptive kernel A type of kernel that allows a process to be preempted while it is running in kernel mode.

preemptive multitasking A model of multitasking in which threads on cores may be preempted by higher-priority threads before finishing their scheduled time quanta.

prepaging In virtual memory, bringing pages into memory before they are requested.

principle of least privilege A design principle stating that every program and every privileged user of the system should operate using the least amount of privilege necessary to complete the job.

priority inversion A scheduling challenge arising when a higher-priority process needs to read

or modify kernel data that are currently being accessed by a lower-priority process.

priority number A number indicating the position of a process in a conditional-wait queue in a monitor construct.

priority paging Prioritizing selection of victim frames based on some criteria, such as avoiding selection of shared library pages.

priority replacement algorithm A virtual memory page replacement algorithm that avoids thrashing by not allowing a process to steal frames from other processes.

priority-inheritance protocol A protocol for solving priority inversion in which all processes that are accessing resources needed by a higher-priority process inherit that higher priority until they are finished with the resources in question.

priority scheduling A scheduling algorithm in which a priority is associated with each thread and the free CPU core is allocated to the thread with the highest priority.

private cloud Cloud computing run by a company for that company's own use.

private key In an asymmetric encryption algorithm, a key that must be kept private for use in authenticating, encrypting, and decrypting.

privilege escalation The enabling of more privileges than an entity (process, system, person) should have.

privileged instructions Instructions that can execute only if the CPU is in kernel mode.

privileged mode A CPU mode in which all instructions are enabled. The kernel runs in this mode. See also user mode.

proc file system (/proc) A pseudo file system using file-system interfaces to provide access to a system's process name space.

procedural language A language that implements state-based algorithms (e.g., C, C++, Java, and C#).

process A program loaded into memory and executing.

process control A category of system calls.

process control block A per-process kernel data structure containing many pieces of information associated with the process.

process identifier (pid) A unique value for each process in the system that can be used as an index to access various attributes of a process within the kernel.

process lifetime management (PLM) A Windows power-saving feature that suspends all

threads within a process that has not been used for a few seconds.

process migration The movement of a process between computers.

process name A human-readable name for a process.

process scheduler A scheduler that selects an available process (possibly from a set of several processes) for execution on a CPU.

process synchronization Coordination of access to data by two or more threads or processes.

process-contention scope (PCS) A scheduling scheme, used in systems implementing the many-to-one and many-to-many threading models, in which competition for the CPU takes place among threads belonging to the same process.

processor affinity A kernel scheduling method in which a process has an affinity for (prefers) the processor on which it is currently running.

processor groups In Windows 7, processors grouped together for management and scheduling.

producer A process role in which the process produces information that is consumed by a consumer process.

production kernels Kernels released for production use (as opposed to development use).

profiling Periodically sampling the instruction pointer to determine which code is being executed; used in debugging and performance tuning.

program counter A CPU register indicating the main memory location of the next instruction to load and execute.

programmable interval timer A hardware timer provided by many CPUs.

programmed I/O (PIO) A method of transferring data between a CPU and a peripheral device in which data are transferred one byte at a time.

programming-environment virtualization Virtualization in which a virtual machine manager does not virtualize real hardware but instead creates an optimized virtual system (examples include Oracle Java and Microsoft.Net).

project In Solaris scheduling, a group of processes scheduled as a unit.

proportional allocation An allocation algorithm that assigns a resource in proportion to some aspect of the requestor. In virtual memory, the assignment of page frames in proportion to the size each process.

proportional share A scheduler that operates by allocating T shares among all applications, ensuring that each gets a specific portion of CPU time.

protection A category of system calls. Any mechanism for controlling the access of processes or users to the resources defined by a computer system.

protection domain In protection, a set of resources that a process may access. In virtualization, a virtual machine manager creates a protection domain for each guest to inform the CPU of which physical memory pages belong to that guest.

protection mask In Linux and UNIX, a set of bits assigned to an object specifying which access modes (read, write, execute) are to be granted to processes with owner, group, or world access writes to the object.

protection rings A model of privilege separation consisting of a series of rings, with each successive ring representing greater execution privileges.

pseudo device driver In virtualization, a guest device driver that does not directly control system hardware but rather works with the virtual machine manager to access the device.

PTE table A Windows virtual-memory data structure.

Pthreads The POSIX standard (IEEE 1003.1c) defining an API for thread creation and synchronization (a specification for thread behavior, not an implementation).

public cloud Cloud computing available via the Internet to anyone willing to pay for the services offered.

public domain The total absence of copyright protection. Software in the public domain can be used as desired by anyone, with no limits.

public key In asymmetric encryption algorithm, a key that can be distributed for encrypting and decrypting.

public key encryption A cipher algorithm in which different keys are used for encryption and decryption.

pull migration Migration that occurs when an idle processor pulls a waiting thread from a busy processor.

pure demand paging A demand paging scheme wherein no page is brought into memory until it is referenced.

push The action of placing a value on a stack data structure.

push migration Migration in which a task periodically checks the load on each processor and, if it finds an imbalance, evenly distributes the load by moving (or pushing) threads from overloaded to idle or less busy processors.

Quest-V An example of a partitioning hypervisor.

queue A sequentially ordered data structure that uses the first-in, first-out (FIFO) principle; items are removed from a queue in the order in which they were inserted.

queueing-network analysis An area of computing study in which algorithms are analyzed for various characteristics and effectiveness.

race condition A situation in which two threads are concurrently trying to change the value of a variable.

RAID levels The various types of RAID protection.

RAM drives Sections of a system's DRAM presented to the rest of the system as if they were secondary storage devices.

random-access memory (RAM) Rewritable memory, also called main memory. Most programs run from RAM, which is managed by the kernel.

ransomware A class of malware that disables computer access (frequently by encrypting files or the entire system) until a ransom is paid.

rate Generally, a measure of speed or frequency. A periodic real-time process has a scheduling rate of $1/p$, where p is the length of its running period.

rate-monotonic A scheduling algorithm that schedules periodic tasks using a static priority policy with preemption.

raw partition A partition within a storage device not containing a file system.

raw disk Direct access to a secondary storage device as an array of blocks with no file system.

raw I/O Direct access to a secondary storage device as an array of blocks with no file system.

read pointer The location in a file from which the next read will occur.

read-ahead Sequential I/O performance optimization that reads and caches several subsequent pages when a read of one page is requested.

readers-writers problem A synchronization problem in which one or more processes or threads write data while others only read data.

reader-writer lock A lock appropriate for access to an item by two types of accessors, read-only and read-write.

read-modify-write cycle The situation in which a write of data smaller than a block requires the entire block to be read, modified, and written back.

read-only memory (ROM) A storage device whose contents are not modifiable.

read-write (RW) Access that allows reading and writing.

ready queue The set of processes ready and waiting to execute.

real-time A term describing an execution environment in which tasks are guaranteed to complete within an agreed-to time.

real-time class A scheduling class that segregates real-time threads from other threads to schedule them separately and provide them with their needed priority.

real-time operating systems (RTOS) Systems used when rigid time requirements have been placed on the operation of a processor or the flow of data; often used as control devices in dedicated applications.

reapers In memory management, routines that scan memory, freeing frames to maintain a minimum level of available free memory.

recovery mode A system boot state providing limited services and designed to enable the system admin to repair system problems and debug system startup.

Red Hat A popular Linux distribution.

red-black tree A tree containing n items and having at most $\lg n$ levels, thus ensuring worst-case performance of $O(\lg n)$.

redirector In Windows, a client-side object that forwards I/O requests to a remote system.

redundant arrays of independent disks (RAID) A disk organization technique in which two or more storage devices work together, usually with protection from device failure.

reentrant code Code that supports multiple concurrent threads (and can thus be shared).

reference bit An MMU bit indicating that a page has been referenced.

reference string A trace of accesses to a resource. In virtual memory, a list of pages accessed over a period of time.

referenced pointer In Windows, a means by which kernel-mode code can access objects; must be obtained by calling a special API.

regions In ARM v8 CPUs, contiguous areas of memory with separate privilege and access rules.

registry A file, set of files, or service used to store and retrieve configuration information. In Windows, the manager of hives of data.

regressive round-robin A variation on round-robin scheduling in which a thread that uses its entire CPU scheduling quantum is given a longer quantum and higher priority.

relative access A file-access method in which contents are read in random order, or at least not sequentially.

relative block number An index relative to the beginning of a file. The first relative block of the file is block 0, the next is block 1, and so on through the end of the file.

relative path name A path name starting at a relative location (such as the current directory).

relocatable code Code with bindings to memory addresses that are changed at loading time to reflect where the code is located in main memory.

relocatable object file The output of a compiler in which the contents can be loaded into any location in physical memory.

relocation An activity associated with linking and loading that assigns final addresses to program parts and adjusts code and data in the program to match those addresses.

relocation register A CPU register whose value is added to every logical address to create a physical address (for primitive memory management).

remainder section Whatever code remains to be processed after the critical and exit sections.

remote access tool (RAT) A back-door daemon left behind after a successful attack to allow continued access by the attacker.

remote desktop The representation of a desktop session to another system across a network, for remote access to the computer's GUI.

remote desktop protocol (RDP) A network protocol to allow remote access to a computer's display contents and keyboard and mouse input devices.

remote file transfer A function of a network operating system providing a means to transfer files between network-attached computers.

remote procedure calls (RPCs) Procedure calls sent across a network to execute on another computer; commonly used in client-server computing.

remote-service mechanism A facility, implemented by a feature such as RPC, in which clients ask a remote system to perform a function for them.

renderer A process that contains logic for rendering contents (such as web pages) onto a display.

rendezvous In interprocess communication, when blocking mode is used, the meeting point at which a send is picked up by a receive.

replay attack The malicious or fraudulent repetition of a valid transmission.

replication In file systems, the duplication and synchronization of a set of data over a network to another system. In storage, the automatic duplication of writes between separate sites.

request edge In a system resource-allocation graph, an edge (arrow) indicating a resource request.

request manager In Linux, the kernel component that manages the reading and writing of buffer contents to and from a block-device driver.

resolve Generally, to translate from a symbolic representation to a numeric one. In networking, to translate from a host name to a host-id. With files, to follow a link and find the target file.

resource allocator An operating system or application that determines how resources are to be used.

resource manager The role of an operating system in managing the computer's resources.

resource sharing The ability for multiple users, computers, etc., to access computing resources.

resource utilization The amount of a given resource (hardware or software) that is being used.

response time The amount of time it takes the system to respond to user action.

restore In file systems, the act of repairing or recovering files or a file system from a backup.

resume In virtualization, the continuation of execution after a guest's suspension.

reverse engineering The procedure of converting a compiled binary file into a human-readable format.

rich text format A file format developed by Microsoft that includes formatting details but can be used by various applications and operating systems, enabling files to be transferred between programs and systems.

risk assessment A systemic security analysis that attempts to value the assets of the entity in question and determine the odds that a security incident will affect the entity.

roaming profile In Windows, a collection of user preferences and settings that are kept on a server and allow a user's environment to follow that user from computer to computer.

role-based access control (RBAC) A method of access control in which roles rather than users have access rights; applies the principle of least privilege to the protection of operating systems.

role In RBAC, a named set of privileges that can be available to a user.

root partition The storage partition that contains the kernel and the root file system; the one mounted at boot.

rotational latency On an HDD, the time it takes the read-write head, once over the desired cylinder, to access the desired track.

round-robin (RR) A scheduling algorithm similar to FCFS scheduling, but with preemption added to enable the system to switch between threads; designed especially for time-sharing systems.

router A device or software that connects networks to each other (e.g., a home network to the Internet).

RSA The most widely used public key cipher.

run queue The queue holding the threads that are ready to run on a CPU.

running The state of the operating system after boot when all kernel initialization has completed and system services have started. In general, the system state after booting and before crashing or being shut down.

run-time environment (RTE) The full suite of software needed to execute applications written in a given programming language, including its compilers, libraries, and loaders.

safe computing Human behavior aimed at avoiding viruses and other security problems (e.g., by avoiding downloading pirated software).

safe sequence “In deadlock avoidance, a sequence of processes $\langle P_1, P_2, \dots, P_n \rangle$ in which, for each P_i , the resource requests that P_i can make can be satisfied by the currently available resources plus the resources held by all P_j , with $j < i$.”

safe state In deadlock avoidance, a state in which a system can allocate resources to each process in some order and still avoid deadlock.

sandbox A contained environment (e.g., a virtual machine).

sandboxing Restricting what an object can do by placing it in a contained environment (e.g., running a process on a virtual machine).

SAS A common type of I/O bus.

scalability Generally, the ability of a facility or service to increase capacity as needed by the users (e.g., to add more cores when the load increases).

SCAN algorithm An HDD I/O scheduling algorithm in which the disk head moves from one end of the disk to the other performing I/O as the head passes the desired cylinders; the head then reverses direction and repeats.

scatter-gather An I/O method in which multiple sources or destinations of I/O are specified in one command structure.

scheduler The part of the operating system that determines the next job to be done (e.g., the next process to be executed).

scheduler activation A threading method in which the kernel provides an application with a set of LWPs, and the application can schedule user threads onto an available virtual processor and receive upcalls from the kernel to be informed of certain events.

scheduling classes In Linux, classes on which scheduling is based; each class is assigned a specific priority.

scheduling domain A set of CPU cores that can be balanced against one another.

scope The time between when a lock is acquired and when it is released.

script kiddie An attacker who did not design the attack but instead is using an attack designed by a more sophisticated attacker.

search path In some operating systems, the sequence of directories searched for an executable file when a command is executed.

second extended file system (ext2) In Linux, an outdated version of the extended file system

secondary storage A storage system capable of holding large amounts of data permanently; most commonly, HDDs and NVM devices.

second-chance page-replacement algorithm A FIFO page replacement algorithm in which, if the reference bit is set, the bit is cleared and the page is not replaced.

second-level interrupt handler In some operating systems, the interrupt handler that actually handles interrupts; reception and queueing of interrupts are handled at another level (by the first-level handler).

section object The Windows data structure that is used to implement shared memory.

sector forwarding The replacement of an unusable HDD sector with another sector at some other location on the device.

sector slipping The renaming of sectors to avoid using a bad sector.

sector sparing The replacement of an unusable HDD sector with another sector at some other location on the device.

sector On an HDD platter, a fixed-size section of a track.

secure The state of a system whose resources are used and accessed as intended under all circumstances.

secure by default Describes a system or computer whose initial configuration decreases its attack surface.

secure monitor call (SMC) An ARM processor special instruction that can be used by the kernel to request services from the TrustZone.

secure system process In Windows, the process representing the fact that the secure kernel is loaded.

security The defense of a system from external and internal attacks. Such attacks include viruses and worms, denial-of-service attacks, identity theft, and theft of service.

security access token In Windows 10, a token created when a user logs in that contains the user's security ID, the security IDs of the groups the user belongs to, and a list of special privileges the user has.

security context In Windows 10, a characteristic, based on a user's access token, that enables a program run by the user to access what the user is allowed to access.

security descriptor In Windows 10, a feature that describes the security attributes of an object.

security domain The separation of systems and devices into classes, with each class having similar security needs.

security ID (SID) In Windows, a value used to uniquely identify a user or group for security purposes.

security policy A document describing the set of things being secured, how they are to be secured, and how users are to behave in matters relating to security.

security reference monitor (SRM) A Windows component that checks the effective security token whenever a thread opens a handle to a protected data structure.

security through obscurity A security layer in which information is kept private or obscured in the hope that it won't be discovered and used by attackers; an ineffective security method.

security token In Windows, a token associated with each process containing the SIDs of the user and the user's groups, the user's privileges, the integrity level of the process, the attributes and claims associated with the user, and any relevant capabilities.

seek The operation of changing the current file-position pointer.

seek time On an HDD, the time it takes the read-write head to position over the desired cylinder.

semaphore An integer variable that, apart from initialization, is accessed only through two standard atomic operations: wait() and signal().

semiconductor memory The various types of memory constructed from semiconductors.

sense key In the SCSI protocol, information in the status register indicating an error.

separation hypervisor An experimental system that uses virtualization to partition separate system components into a chip-level distributed computing system.

sequence number In networking, a counter assigned to packets to order their assembly after delivery.

sequential access A file-access method in which contents are read in order, from beginning to end.

serial-attached SCSI (SAS) A common type of I/O bus.

server In general, any computer, no matter the size, that provides resources to other computers.

server subject In Windows 10 security, a process implemented as a protected server that uses the security context of the client when acting on the client's behalf.

server system A system providing services to other computers (e.g., a web server).

server-message-block (SMB) The Windows protocol for sending I/O requests over a network; a version was published as the common internet file system (CIFS).

service A software entity running on one or more machines and providing a particular type of function to calling clients. In Android, an application component with no user interface; it runs in the background while executing long-running operations or performing work for remote processes.

service control manager (SCM) In Windows 7, the component that manages services associated with plug-and-play devices.

service-trigger A mechanism in Windows 7 that allows plug-and-play device insertion to launch a service.

session In networking, a complete round of communication, frequently beginning with a login and ending with a logoff to terminate communications.

session hijacking The interception of a communication.

session key The TLS symmetric key, used for a web communication session, exchanged via asymmetric cryptography.

SHA-1 An algorithm for creating a hash (a small, fixed-size block of data calculated from a larger data set, used to determine if a message has been changed).

shared libraries Libraries that can be loaded into memory once and used by many processes; used in systems that support dynamic linking.

shared lock A file lock similar to a reader lock in that several processes can obtain the lock concurrently.

shared memory In interprocess communication, a section of memory shared by multiple processes and used for message passing.

shared system interconnect A bus connecting CPUs to memory in such a way that all CPUs can access all system memory; the basis for NUMA systems.

shared-memory model An interprocess communication method in which multiple processes share memory and use that memory for message passing.

shares A basis for making scheduling decisions. The fair-share scheduling class uses CPU shares instead of priorities to allocate CPU time.

shell One of the command interpreters on a system with multiple command interpreters to choose from.

shell script A file containing a set series of commands (similar to a batch file) that are specific to the shell being used.

shortest-job-first (SJF) A scheduling algorithm that associates with each thread the length of the thread's next CPU burst and schedules the shortest first.

shortest-remaining-time-first (SJRF) A scheduling algorithm that gives priority to the thread with the shortest remaining time until completion.

shortest-seek-time-first (SSTF) algorithm An HDD I/O scheduling algorithm that sorts requests by the amount of seek time required to accomplish the request; the shortest time has the highest priority.

shoulder surfing Attempting to learn a password or other secret information by watching the target user at the keyboard.

siblings In a tree data structure, child nodes of the same parent.

Siemens Jailhouse An example of a partitioning hypervisor.

signal In UNIX and other operating systems, a means used to notify a process that an event has occurred.

signature In intrusion detection, a pattern of behavior associated with an attack.

simple subject In Windows 10 security, a subject that manages a user-initiated program's permissions.

simultaneous multithreading (SMT) The situation in which, in a CPU with multiple cores, each core supports multiple hardware threads.

single indirect block In UFS, a block containing pointers to direct blocks, which point to data blocks.

single instruction multiple data (SIMD) A form of parallelism in which multiple compute elements perform the same single instruction operating on multiple data points.

single step A CPU mode in which a trap is executed by the CPU after every instruction (to allow examination of the system state after every instruction); useful in debugging.

single-threaded A process or program that has only one thread of control (and so executes on only one core at a time).

single-user mode A system boot state providing limited services and designed to enable the system admin to repair system problems and debug system startup.

Siri The Apple voice-recognition system.

sketch An Arduino program.

slab A section of memory made up of one or more contiguous pages; used in slab allocation.

slab allocation A memory allocation method in which a slab of memory is allocated and split into chunks that hold objects of a given size. As the objects are freed, the chunks can coalesce into larger chunks, eliminating fragmentation.

Slackware An early but still widely used Linux distribution.

slim reader-write lock (SRW) A type of lock in modern Windows OS that favors neither readers nor writers.

small computer-systems interface (SCSI) One type of interface between a system and its storage (SCSI). See also ATA and SATA.

snapshot In file systems, a read-only view of a file system at a particular point in time; later changes do not affect the snapshot view.

sniff In network communication, to capture information by recording data as it is transmitted.

sniffing An attack in which the attacker monitors network traffic to obtain useful information.

social engineering A practice in which an attacker tricks someone into performing some task for the attacker (such as sending the attacker confidential information).

socket An endpoint for communication. An interface for network I/O.

soft affinity An operating system's policy of attempting to keep a process running on the same processor but not guaranteeing that it will do so.

soft error An error that is recoverable by retrying the operation.

soft real-time systems Systems that provide no guarantee as to when a critical real-time thread will be scheduled; they guarantee only that the thread will be given preference over noncritical threads

Software as a Service (SaaS) A type of computing in which one or more applications (such as word processors or spreadsheets) are available as a service via the Internet.

software engineering A field of study and a career involving writing software (i.e., programming.)

software interrupt A software-generated interrupt; also called a trap. The interrupt can be caused either by an error (e.g., division by zero or invalid memory access) or by a specific request from a user program that an operating-system service be performed.

software objects The software components that make up a computer or device (files, programs, semaphores, etc.).

software transactional memory (STM) Transactional memory implemented exclusively in software; no special hardware is needed.

Solaris A UNIX derivative that is the main operating system of Sun Microsystems (now owned by Oracle Corporation). There is an active open source version called Illumos.

Solaris ZFS An advanced file system, first included as part of Solaris.

solid-state disk A disk-drive-like storage device that uses flash-memory-based nonvolatile memory.

source file A file containing the source code of a program.

space sharing A feature of APFS in which storage is treated as a pool and space is shared among the file systems created in that pool (much like ZFS).

SPARC A proprietary RISC CPU created by Sun Microsystems and now owned by Oracle Corporation. There is an active open source version called OpenSPARC.

sparse In memory management, a term describing a page table that has noncontiguous, scattered entries. A sparse address space has many holes.

spinlock A locking mechanism that continuously uses the CPU while waiting for access to the lock.

split-screen Running multiple foreground processes (e.g., on an iPad) but splitting the screen among the processes.

spoof The imitation of a legitimate identifier (such as an IP address) by an illegitimate user or system.

spool A buffer that holds output for a device (such as a printer) that cannot accept interleaved data streams.

springboard The iOS touch-screen interface.

spyware A Trojan horse variation in which the installed malware gathers information about a person or organization.

stack A sequentially ordered data structure that uses the last-in, first-out (LIFO) principle for adding and removing items; the last item placed onto a stack is the first item removed.

stack algorithm A class of page-replacement algorithms that do not suffer from Belady's anomaly.

stack inspection In Java, a protection procedure in which a calling sequence is checked to ensure that some caller in the sequence has been granted access to the resource called.

stack section The section of process memory that contains the stack; it contains activation records and other temporary data.

stall A CPU state occurring when the CPU is waiting for data from main memory and must delay execution.

starvation The situation in which a process or thread waits indefinitely within a semaphore. Also, a scheduling risk in which a thread that is ready to run is never put onto the CPU due to the scheduling algorithm; it is starved for CPU time.

state The condition of a process, including its current activity as well as its associated memory and disk contents.

state information In remote file systems, the set of information pertaining to connections and ongoing file operations (e.g., which files are open).

state restore Copying a process's context from its saved location to the CPU registers in preparation for continuing the process's execution.

state save Copying a process's context to save its state in order to pause its execution in preparation for putting another process on the CPU.

stateless In remote file systems, a protocol in which state need not be maintained for proper operation.

static linking Linking in which system libraries are treated like other object modules and combined by the loader into a binary program image.

status register A device I/O register in which status is indicated.

storage-area network (SAN) A local-area storage network allowing multiple computers to connect to one or more storage devices.

stream cipher A cipher that encrypts or decrypts a stream of bits or bytes (rather than a block).

stream head The interface between STREAMS and user processes.

stream modules In STREAMS, modules of functionality loadable into a STREAM.

STREAMS A UNIX I/O feature allowing the dynamic assembly of pipelines of driver code.

stub A small, temporary place-holder function replaced by the full function once its expected behavior is known.

subject In Windows 10 security, an entity used to track and manage user permissions.

subsystem A subset of an operating system responsible for a specific function (e.g., memory management).

SunOS The predecessor of Solaris by Sun Microsystems Inc.

superblock The UFS volume control block.

superblock object The VFS representation of the entire file system.

supervisor mode A CPU mode in which all instructions are enabled. The kernel runs in this mode. See also user mode.

SuSE A popular Linux distribution.

suspend In virtualization, to freeze a guest operating system and its applications to pause execution.

swap map In Linux swap-space management, a part of the data structure tracking swap-space use.

swap space Secondary storage backing-store space used to store pages that are paged out of memory.

swapped Moved between main memory and a backing store. A process may be swapped out to free main memory temporarily and then swapped back in to continue execution.

swapping Moving a process between main memory and a backing store. A process may be swapped out to free main memory temporarily and then swapped back in to continue execution.

swap-space management The low-level operating-system task of managing space on secondary storage for use in swapping and paging.

symmetric clustering A situation in which two or more hosts are running applications and are monitoring each other.

symmetric encryption algorithm A cryptography algorithm in which the same keys are used to encrypt and decrypt the message or data.

symmetric multiprocessing (SMP) Multiprocessing in which each processor performs all tasks, including operating-system tasks and user processes. Also, a multiprocessor scheduling method in which each processor is self-scheduling and may run kernel threads or user-level threads.

synchronous In interprocess communication, a mode of communication in which the sending process is blocked until the message is received by the receiving process or by a mailbox and the receiver blocks until a message is available. In I/O, a request that does not return until the I/O completes.

synchronous threading Threading in which a parent thread creating one or more child threads waits for them to terminate before it resumes.

synchronous writes Writes that are stored in the order in which they were issued, are not buffered, and have requesting threads wait for the writes to complete before continuing.

system administrators Computer users that configure, monitor, and manage systems.

system build Creation of an operating-system build and configuration for a specific computer site.

system call Software-triggered interrupt allowing a process to request a kernel service.

system call The primary interface between processes and the operating system, providing a means to invoke services made available by the operating system.

system-call filtering An operating-system facility to limit which system calls can be executed by a process.

system daemon A service that is provided outside the kernel by system programs that are loaded into memory at boot time and run continuously.

system disk A storage device that has a boot partition and can store an operating system and other information for booting the computer.

system integrity protection (SIP) A feature of macOS 10.11 and later versions that uses extended file attributes to mark system files as restricted so that even the root user cannot tamper with them.

system mode A CPU mode in which all instructions are enabled. The kernel runs in this mode. See also user mode.

system process A service that is provided outside the kernel by system programs that are loaded into memory at boot time and run continuously. In

Windows, a process that serves as the container of all internal kernel worker threads and other system threads created by drivers for polling, house-keeping, and other background work.

system program A program associated with the operating system but not necessarily part of the kernel.

system resource-allocation graph A directed graph for precise description of deadlocks.

system restore point In Windows, a copy of the system hives taken before any significant change is made to system configuration.

system service A collection of applications included with or added to an operating system to provide services beyond those provided by the kernel.

system utility A collection of applications included with or added to an operating system to provide services beyond what are provided by the kernel.

system-call firewall A firewall within a computer that limits the system calls a process can trigger.

system-call interface An interface that serves as the link to system calls made available by the operating system and that is called by processes to invoke system calls.

system-contention scope (SCS) A thread-scheduling method in which kernel-level threads are scheduled onto a CPU regardless of which process they are associated with (and thus contend with all other threads on the system for CPU time).

system-development time The time during which an operating system is developed, before it is made available in final “release” form.

system-wide open-file table A kernel in-memory data structure containing a copy of the FCB of each open file, as well as other information.

target latency In the Completely Fair Scheduler, a configurable variable which is the interval of time during which every runnable task should run at least once.

targeted latency An interval of time during which every runnable thread should run at least once.

task control block A per-process kernel data structure containing many pieces of information associated with the process.

task parallelism A computing method that distributes tasks (threads) across multiple computing cores, with each task is performing a unique operation.

task A process, a thread activity, or, generally, a unit of computation on a computer.

templating In virtualization, using one standard virtual-machine image as a source for multiple virtual machines.

terabyte (TB) 1,024⁴ bytes.

terminal concentrator A type of front-end processor for terminals.

tertiary storage A type of storage that is slower and cheaper than main memory or secondary storage; frequently magnetic tape or optical disk.

text file A type of file containing text (alphanumeric characters).

text section The executable code of a program or process.

thin client A limited computer (terminal) used for web-based computing.

third extended file system (ext3) In Linux, a current version of the extended file system; the successor to ext2.

thrashing Paging memory at a high rate. A system thrashes when there is insufficient physical memory to meet virtual memory demand.

thread A process control structure that is an execution location. A process with a single thread executes only one task at a time, while a multi-threaded process can execute a task per thread.

thread cancellation Termination of a thread before it has completed.

thread dump In Java, a snapshot of the state of all threads in an application; a useful debugging tool for deadlocks.

thread library A programming library that provides programmers with an API for creating and managing threads.

thread pool A number of threads created at process startup and placed in a pool, where they sit and wait for work.

thread-environment block (TEB) In Win32, a user-mode threads data structure that contains numerous per-thread fields.

thread-local storage (TLS) Data available only to a given thread.

threat The potential for a security violation.

throughput Generally, the amount of work done over time. In scheduling, the number of threads completed per unit time.

tightly coupled systems Systems with two or more processors in close communication, sharing the computer bus and sometimes the clock, memory, and peripheral devices.

time quantum A small unit of time used by scheduling algorithms as a basis for determining when to preempt a thread from the CPU to allow another to run.

time sharing A practice in which the CPU executes multiple jobs by switching among them, but the switches occur so frequently that the users can interact with the processes.

time slice A small unit of time used by scheduling algorithms as a basis for determining when to preempt a thread from the CPU to allow another to run.

timer A hardware component that can be set to interrupt the computer after a specified period.

timestamp counter (TSC) In Windows Vista, a counter that tracks execution time.

TLB miss A translation look-aside buffer lookup that fails to provide the address translation because it is not in the TLB.

TLB reach The amount of memory addressable by the translation look-aside buffer.

TLB walk The steps involved in walking through page-table structures to locate the needed translation and then copying that result into the TLB.

touch screen A touch-sensitive screen used as a computer input device.

touch-screen interface A user interface in which touching a screen allows the user to interact with the computer.

trace tapes A tool used in the evaluation of scheduling algorithms. Thread details are captured on real systems, and various algorithms are analyzed to determine their effectiveness.

track On an HDD platter, the medium that is under the read-write head during a rotation of the platter.

transaction Generally, the execution of a set of steps that make up one activity. In log-based transaction-oriented file systems, a set of operations completed as part of a request (e.g., “write this block to that file”).

transactional memory A type of memory supporting memory transactions.

transfer rate The rate at which data flows.

translation granules Features of ARM v8 CPUs that define page sizes and regions.

translation look-aside buffer (TLB) A small, fast-lookup hardware cache used in paged memory address translation to provide fast access to a subset of memory addresses.

translation table base register ARM v8 CPU register pointing to the level 0 (outer) page table for the current thread.

transmission control protocol/Internet protocol (TCP/IP) The most common network protocol; it provides the fundamental architecture of the Internet.

transparent In distributed computing, a term describing the automatic sharing of resources so that users do not know if a resource is local or remote.

transport driver interface (TDI) In Windows networking, an interface that supports connect-based and connectionless transports on top of the transport layer.

transport layer security (TLS) A cryptographic protocol that enables two computers to communicate securely; the standard protocol by which web browsers communicate to web servers.

trap A software interrupt. The interrupt can be caused either by an error (e.g., division by zero or invalid memory access) or by a specific request from a user program that an operating-system service be performed.

trap door A back-door daemon left behind after a successful attack to allow continued access by the attacker.

trap-and-emulate In virtualization, a method used to implement virtualization on systems lacking hardware support (such as CPU instructions) for virtualization; any action that would cause the guest to call the operating system is intercepted, and the result is emulated.

tree A data structure that can be used to represent data hierarchically; data values in a tree structure are linked through parent-child relationships.

triple DES A modification of DES that uses the same algorithm three times and uses two or three keys to make the encryption more difficult to break.

triple indirect block In UFS, a block containing pointers to double indirect blocks, which point to single indirect blocks, which point to data blocks.

Trojan horse A program that acts in a clandestine or malicious manner rather than simply performing its stated function.

TrustZone (TZ) ARM processor implementation of the most secure protection ring.

tunnel In computer communication, a container of communications within another type of communication (e.g., a VPN that allows web traffic).

turnstile A Solaris scheduling feature using a queue structure containing threads blocked on a lock.

two-factor authentication Authentication based on two separate sources of data (e.g., a brain providing a password and a finger providing a fingerprint).

type 0 hypervisor A hardware-based virtualization solution that provides support for virtual machine creation and management via firmware (e.g., IBM LPARs and Oracle LDOMs).

type 1 hypervisor Operating-system-like software built to provide virtualization (e.g., VMware ESX, Joyent SmartOS and Citrix XenServer).

type 2 hypervisor An application that runs on standard operating systems but provides virtual machine management features to guest operating systems (e.g., VMware Workstation and Fusion, and Oracle VirtualBox).

type safety In Java, a feature that ensures that classes cannot treat integers as pointers, write past the end of an array, or otherwise access memory in arbitrary ways.

unbounded buffer A buffer with no practical limit on its memory size.

uncontended A term describing a lock that is available when a thread attempts to acquire it.

unified buffer cache In file I/O, a cache used for both memory-mapped I/O and direct file I/O.

unified extensible firmware interface (UEFI) The modern replacement for BIOS containing a complete boot manager.

unified virtual memory In file I/O, the use of page caching for all types of I/O (explicit file system I/O and page fault I/O).

uniform memory access (UMA) Access to all main memory by all processors, without performance differences based on CPU or memory location.

uniform naming convention (UNC) A name format that includes the system and its resources (e.g. `m\\server_name\share_name\x\y\z`).

unikernels Specialized machine images that contain both an operating system and applications for efficient execution and increased security.

universal serial bus (USB) A type of I/O bus.

universal Windows platform (UWP) Windows 10 architecture that provides a common app platform for all devices that run it, including mobile devices.

UNIX file system (UFS) An early UNIX file system; uses inodes for FCB.

UnixBSD A UNIX derivative based on work done at the University of California at Berkeley (UCB).

unnamed semaphore A POSIX scheduling construct that can only be used by threads in the same process.

unstructured data Data that are not in a fixed format (like a database record) but rather are free-form (like a twitter.com tweet).

upcall A threading method in which the kernel sends a signal to a process thread to communicate an event.

upcall handler A function in a process that handles upcalls.

USB drive Nonvolatile memory in the form of a device that plugs into a USB port.

user The human using a computer, or the identification of the human to the computer.

user account In Windows 10, an account belonging to a user (rather than a system account used by the computer).

user authentication The identification of a user of a computer.

user datagram protocol (UDP) A communications protocol layered on IP that is connectionless, is low latency, and does not guarantee delivery.

user experience layer In the layered macOS and iOS operating system design, the layer that defines the software interface that allows users to interact with computing devices.

user file directory (UFD) In two-level directory implementation, a per-user directory of files.

user identifier (user ID) (UID) A unique numerical user identifier.

user interface (UI) A method by which a user interacts with a computer.

user mode A CPU mode for executing user processes in which some instructions are limited or not allowed. See also kernel mode.

user programs User-level programs, as opposed to system programs.

user rights Permissions granted to users.

user thread A thread running in user mode.

user-defined signal handler The signal handler created by a process to provide non-default signal handling.

user-initiated In the Grand Central Dispatch Apple OS scheduler, the scheduling class representing tasks that interact with the user but need longer processing times than user-interactive tasks.

user-interactive In the Grand Central Dispatch Apple OS scheduler, the scheduling class representing tasks that interact with the user.

user-mode driver framework (UMDF) A framework in Windows to facilitate the writing of user-mode device drivers.

user-mode scheduling (UMS) A Microsoft Windows 7 feature that allows applications to create and manage threads independently of the kernel. This feature supports task-based parallelism by

decomposing processes into tasks, which are then scheduled on available CPUs; it is used on AMD64 systems.

user-mode thread (UT) In Windows, the state of a thread when it is running in user mode.

utility In the Grand Central Dispatch Apple OS scheduler, the scheduling class representing tasks that require a longer time to complete but do not demand immediate results.

utility storage An inServ feature in which storage space can be increased as needed.

valid-invalid A page-table bit indicating whether a page-table entry points to a page within the logical address space of that process.

variable-partition A simple memory-allocation scheme in which each partition of memory contains exactly one process.

vectored I/O An I/O method in which multiple sources or destinations of I/O are specified in one command structure.

version control system Software that manages software distributions by allowing contributors to “push” changes into a repository and “pull” a version of the software source-code tree to a system (e.g., for compilation).

victim frame In virtual memory, the frame selected by the page-replacement algorithm to be replaced.

view In Windows, an address range mapped in shared memory. Also, the second step in memory-mapping a file, allowing a process to access the file contents.

virtual address An address generated by the CPU; must be translated to a physical address before it is used.

virtual address control block (VACB) The data structure in Windows that represents a cache block in the unified I/O cache.

virtual address descriptor (VAD) In Windows, a per-process descriptor of a virtual address range, kept in a tree data structure.

virtual address space The logical view of how a process is stored in memory.

virtual CPU (VCPU) In virtualization, a virtualized host CPU available to allocate to a guest operating system by the virtual machine manager.

virtual file system (VFS) The file-system implementation layer responsible for separating file-system-generic operations and their implementation and representing a file throughout a network

virtual machine (VM) The abstraction of hardware allowing a virtual computer to execute on a

physical computer. Multiple virtual machines can run on a single physical machine (and each can have a different operating system).

virtual machine control structures (VMCSs) Hardware features provided by CPUs that support virtualization to track guest state.

virtual machine manager (VMM) The computer function that manages the virtual machine; also called a hypervisor.

virtual machine sprawl The situation in which there are so many virtual machines on a system that their use, history, and state become confusing and difficult to track; caused by the ease of creating virtual machines.

virtual memory A technique that allows the execution of a process that is not completely in memory. Also, separation of computer memory address space from physical into logical, allowing easier programming and larger name space.

virtual memory fork The `vfork()` system call, which forks a child process, suspends the parent, and lets the child share the parent’s address space for both read and write operations (changes are visible to the parent).

virtual private network (VPN) An encrypted tunnel between two systems, commonly using IPSec, allowing secure remote access.

virtual run time A Linux scheduling aspect that records how long each task has run by maintaining the virtual run time of each task.

virtual trust level (VTL) A Windows 10 virtualization feature using Hyper-V to add more secure system modes.

virtualization A technology for abstracting the hardware of a single computer into several different execution environments, thereby creating the illusion that each environment is running on its own private computer.

virtual-to-physical (V-to-P) In virtualization, the conversion of a virtual machine guest to a physical system’s operating system and applications.

virus A fragment of code embedded in a legitimate program that, when executed, can replicate itself; may modify or destroy files and cause system crashes and program malfunctions.

virus dropper The part of a virus that inserts the virus into the system.

virus signature A pattern that can be used to identify a virus within a system.

VMware Virtualization software company.

VMware Workstation A popular commercial type 2 hypervisor for x86 Windows systems.

vnode The virtual file system file representation structure, similar to the FCB for local files but applied to remote files.

voice recognition A computer interface based on spoken commands, which the computer parses and turns into actions.

volatile Describes storage whose content can be lost in a power outage or similar event.

volatile storage Storage whose content can be lost in a power outage or similar event.

volume A container of storage; frequently, a device containing a mountable file system (including a file containing an image of the contents of a device).

volume control block A per-volume storage block containing data describing the volume.

von Neumann architecture The structure of most computers, in which both process instructions and data are stored in the same main memory.

VT-x Intel x86 CPU virtualization-supporting instructions.

wait queue In process scheduling, a queue holding processes waiting for an event to occur before they need to be put on CPU.

wait set In Java, a set of threads, each waiting for a condition that will allow it to continue.

wait-for graph In deadlock detection, a variant of the resource-allocation graph with resource nodes removed; indicates a deadlock if the graph contains a cycle

wear leveling In nonvolatile memory, the effort to select all NAND cells over time as write targets to avoid premature media failure due to wearing out a subset of cells.

wide-area network (WAN) A network that links buildings, cities, or countries.

WiFi Wireless networking, consisting of devices and protocols that allow devices to attach to a network via radio waves rather than cables.

Win32 API The fundamental interface to the capabilities of Windows.

Windows 10 A release of Microsoft Windows from 2009.

Windows group policy In Windows, a policy providing centralized management and configuration of operating systems, applications, and user settings in an Active Directory environment.

Windows Subsystem for Linux (WSL) A Windows 10 component allowing native Linux applications (ELF binaries) to run on Windows.

Windows XP A widely popular version of Microsoft Windows released in 2001.

Winsock The Windows socket API (similar to BSD sockets) for network communications.

wired down A term describing a TLB entry that is locked into the TLB and not replaceable by the usual replacement algorithm.

wireless network A communication network composed of radio signals rather than physical wires.

witness A lock order verifier.

word A unit made up of one or more bytes. For example, a computer that has 64-bit registers and 64-bit memory addressing typically has 64-bit (8-byte) words.

working set The set of pages in the most recent page references.

working-set maximum The maximum number of frames allowed to a process in Windows.

working-set minimum The minimum number of frames guaranteed to a process in Windows.

working-set model A model of memory access based on tracking the set of most recently accessed pages.

working-set window A limited set of most recently accessed pages (a “window” view of the entire set of accessed pages).

workstation A powerful personal computer (PC) for engineering and other demanding workloads.

world rights A category of file access rights.

World Wide Web (WWW) The Internet; a worldwide system of interconnected computer networks.

WORM Write-once, read-many-times storage.

worm A program that spreads malware between computers without intervention from humans.

worst-fit In memory allocation, selecting the largest hole available.

write amplification The creation of I/O requests not by applications but by NVM devices doing garbage collection and space management, potentially impacting the devices’ write performance.

write pointer The location in a file to which the next write will occur.

write-anywhere file layout (WAFL) The file system that is the heart of the NetApp, Inc., storage appliances.

write-back caching In caching, a policy whereby data are first written to the cache; later, the cache writes the change to the master copy of the data.

write-on-close policy In caching, a policy whereby writes reside in the cache until the file is

closed and are only then written back to the master copy of the data.

write-through policy In caching, a policy whereby writes are not cached but are written through the cache to the master copy of the data.

x86-64 A class of 64-bit CPUs running an identical instruction set; the most common CPUs in desktop and server systems.

Xen Virtualization software company.

XML firewall A firewall that examines and limits XML traffic.

Xtratum An example of a partitioning hypervisor.

yellow pages A distributed naming service that provides username, password, hostname, and printer information to a set of computers.

zero-day attacks Attacks that have not been seen before and therefore cannot be detected via their signatures.

zero-fill-on-demand The writing of zeros into a page before it is made available to a process (to keep any old data from being available to the process).

ZFS Oracle file system, created by Sun Microsystems, with modern algorithms and features and few limits on file and device sizes.

zombie A process that has terminated but whose parent has not yet called wait() to collect its state and accounting information.

zombie systems Compromised systems that are being used by attackers without the owners' knowledge.

zones In application containment, a virtual layer between the operating system and a process in which the application runs, limiting its normal access to system resources. In Linux, the four regions of kernel memory.

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